Physics 9702 Paper 4 · Topic 12

Motion in a circle

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/42 October/November 2021, Question 1 (8 marks) page 2

Examiner report: (ii) Candidates found this question difficult and many demonstrated a weak understanding of the forces involved in circular motion. There was a widespread misunderstanding that centripetal force is some sort of additional force that has to 'balance' the weight; very few candidates appeared to understand that the centripetal force is the resultant of the weight and the normal contact force from the track.

2 9702/41 May/June 2023, Question 2 (12 marks) page 4

Examiner report: (a) This question proved to be challenging. (ii) This 'show that' question proved to be challenging as candidates found it difficult to resolve forces correctly here.

3 9702/42 October/November 2025, Question 1 (9 marks) page 6

Examiner report: Very few candidates were able to deduce that the other force is the contact force from the surface of the Earth upwards from the surface and slightly to the left.

4 9702/42 October/November 2023, Question 1 (10 marks) page 8

Examiner report: (ii) This question proved to be a little more challenging for some of the candidates, but stronger candidates were generally able to calculate the value of L correctly. (d) This was a challenging question.

5 9702/42 February/March 2025, Question 1 (9 marks) page 10

Examiner report: Candidates found it more difficult to use the trigonometry involved in getting to the magnitude of the resultant force from the weight of the sphere. (e) A very common misconception was that the angular frequency must remain unchanged, leading to direct proportionality between speed and radius.

6 9702/41 October/November 2025, Question 1 (15 marks) page 12

Examiner report: It was rarely written that the magnitude of the acceleration was also constant. Only the strongest candidates appreciated that drawing the shadow as a whole gave a greater opportunity to correctly identify the position of maximum positive acceleration.

7 9702/42 May/June 2026, Question 1 (10 marks) page 15

Examiner report: No examiner report yet (May/June 2026). Chosen on judgement: resolving the tilted lift force into a vertical component that balances the weight and a horizontal component that provides the centripetal force, then using $\tan\theta = v^2/rg$, is a step many candidates miss.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

1 **(a)** State what is meant by *centripetal* acceleration.

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.....

..... [1]

(b) An unpowered toy car moves freely along a smooth track that is initially horizontal. The track contains a vertical circular loop around which the car travels, as shown in Fig. 1.1.

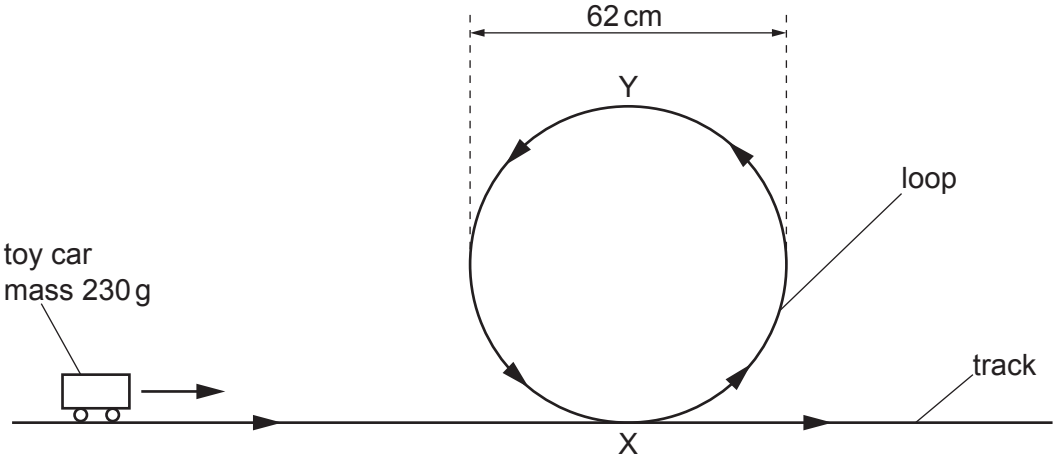


Fig. 1.1

The mass of the car is 230 g and the diameter of the loop is 62 cm. Assume that the resistive forces acting on the car are negligible.

(i) State what happens to the magnitude of the centripetal acceleration of the car as it moves around the loop from X to Y.

..... [1]

(ii) Explain, if the car remains in contact with the track, why the centripetal acceleration of the car at point Y must be greater than $9.8\,m\,s^{-2}$.

.....

.....

..... [2]

(c) The initial speed at which the car in **(b)** moves along the track is $3.8\,m\,s^{-1}$.

Determine whether the car is in contact with the track at point Y. Show your working.

[3]

(d) Suggest, with a reason but without calculation, whether your conclusion in **(c)** would be different for a car of mass 460 g moving with the same initial speed.

.....

.....

..... [1]

[Total: 8]

2 A steel sphere of mass 0.29 kg is suspended in equilibrium from a vertical spring. The centre of the sphere is 8.5 cm from the top of the spring, as shown in Fig. 2.1.

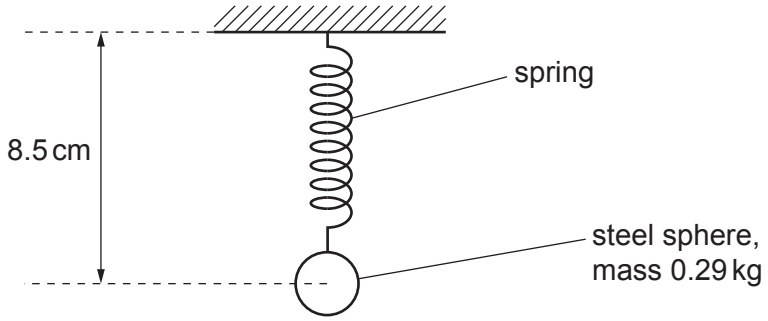


Fig. 2.1

The sphere is now set in motion so that it is moving in a horizontal circle at constant speed, as shown in Fig. 2.2.

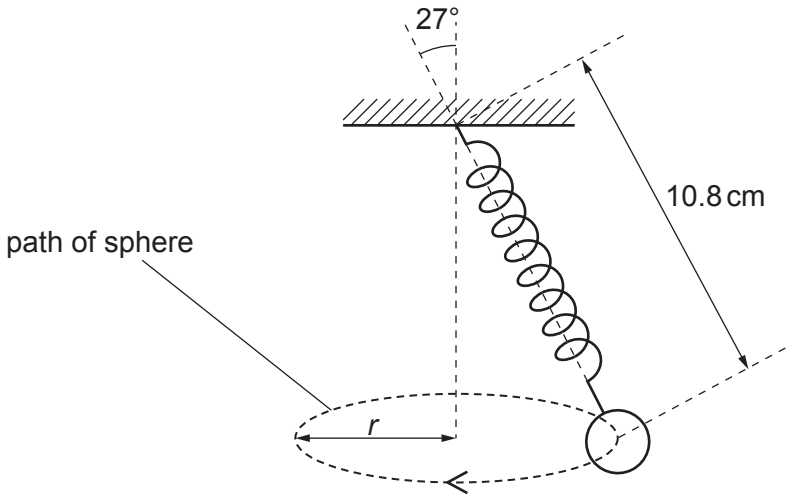


Fig. 2.2

The distance from the centre of the sphere to the top of the spring is now 10.8 cm.

(a) Explain, with reference to the forces acting on the sphere, why the length of the spring in Fig. 2.2 is greater than in Fig. 2.1.

.....

.....

.....

.....

..... [3]

(b) The angle between the linear axis of the spring and the vertical is 27°.

(i) Show that the radius r of the circle is 4.9 cm.

[1]

(ii) Show that the tension in the spring is 3.2 N.

[2]

(iii) The spring obeys Hooke's law.

Calculate the spring constant, in N cm^{-1}, of the spring.

spring constant = ... N cm^{-1} [2]

(c) **(i)** Use the information in **(b)** to determine the centripetal acceleration of the sphere.

centripetal acceleration = ... m s^{-2} [2]

(ii) Calculate the period of the circular motion of the sphere.

period = ... s [2]

[Total: 12]

1 The Earth may be considered as a uniform sphere of radius 6.37×10^6 m.

Cambridge is at a point on the Earth's surface that has a latitude of 52.2° north of the Equator, as shown in Fig. 1.1.

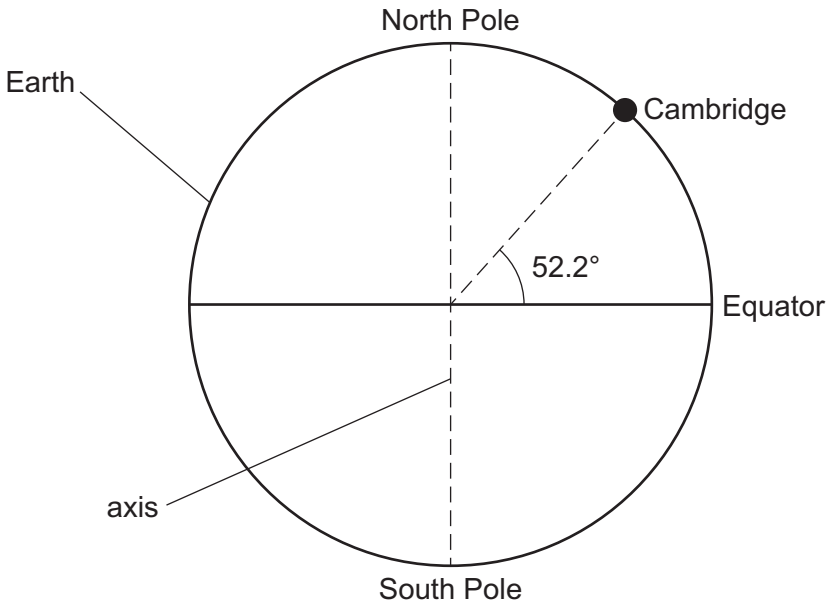


Fig. 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

(a) (i) Show that the radius of the circle around which Cambridge moves is 3.90×10^6 m.

[1]

(ii) Calculate the speed at which Cambridge moves around the circle.

speed = ... $m\,s^{-1}$ [3]

(b) A student of mass 58.6 kg stands on horizontal ground in Cambridge.

(i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = ... N [2]

(ii) On Fig. 1.2, draw an arrow to show the direction of the resultant force that acts on the student.

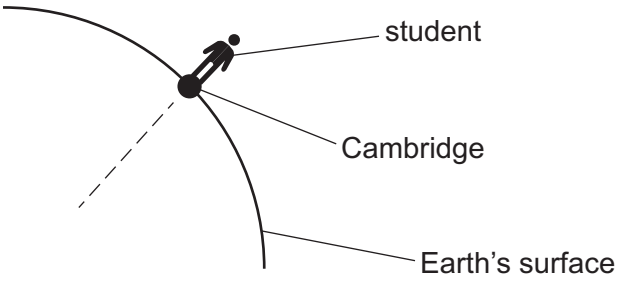


Fig. 1.2 (not to scale)

[1]

(iii) On Fig. 1.3, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in **(b)(ii)**.

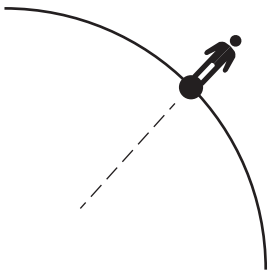


Fig. 1.3 (not to scale)

[2]

[Total: 9]

1 **(a)** Define the radian.

.....

..... [1]

(b) The minute hand of a clock revolves at constant angular speed around the face of the clock, completing one revolution every hour. A small piece of modelling clay is attached to the hand with its centre of gravity at a distance *L* from the fixed end of the hand, as shown in Fig. 1.1.

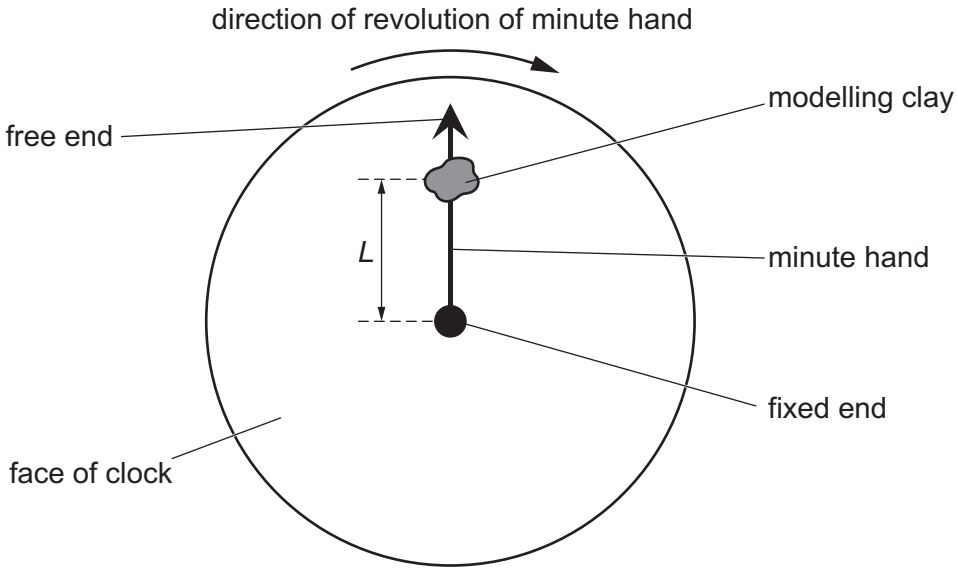


Fig. 1.1

Calculate the angular speed ω of the minute hand.

ω = ... rad s^{-1} [2]

(c) During a time interval of 1400 s, the centre of gravity of the piece of modelling clay in Fig. 1.1 moves through a total distance of 0.44 m.

(i) Calculate the angle through which the minute hand moves in this time interval.

angle = ... rad [1]

(ii) Determine distance L.

L = ... m [2]

(iii) Calculate the magnitude of the centripetal acceleration of the piece of modelling clay.

centripetal acceleration = ... $m\,s^{-2}$ [2]

(d) Use your answer in **(c)(iii)** to explain why the variation with time of the magnitude of the force exerted by the minute hand on the piece of modelling clay is negligible as the minute hand undergoes one full revolution.

.....

.....

..... [2]

[Total: 10]

1 A steel ball is placed on the inside surface of a hollow circular cone. The ball moves in a horizontal circle at constant speed, as shown in Fig. 1.1.

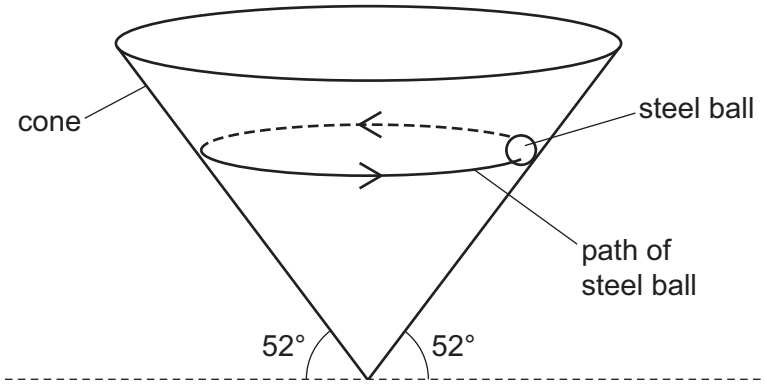


Fig. 1.1

The angle of the side of the cone to the horizontal is 52°. There is no friction between the ball and the cone.

(a) Fig. 1.2 shows a cross-section through the cone and the steel ball.

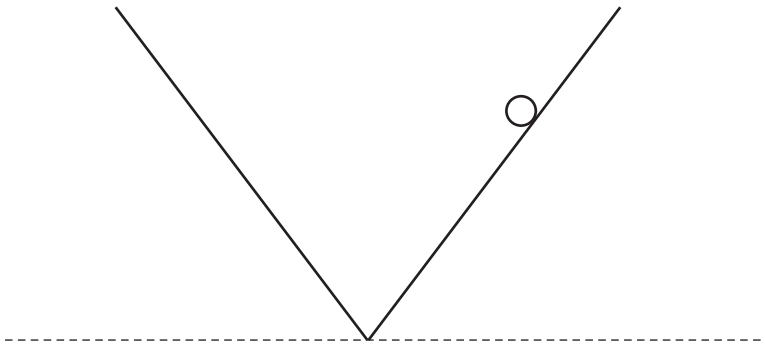


Fig. 1.2

On Fig. 1.2, draw labelled arrows to show the **two** forces acting on the ball. [1]

(b) Describe how the forces acting on the ball cause its acceleration to be centripetal.

.....

.....

.....

..... [2]

(c) The ball moves in a circle of radius 0.15 m.

Show that the speed of the ball is 1.4 m s^{-1}.

[3]

(d) Calculate the angular speed ω of the ball.

ω = ... rad s^{-1} [2]

(e) The speed of the ball is increased.

Explain why the radius of the circular path of the ball increases.

.....

.....

..... [1]

[Total: 9]

1 **(a)** In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

(b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius *R*.

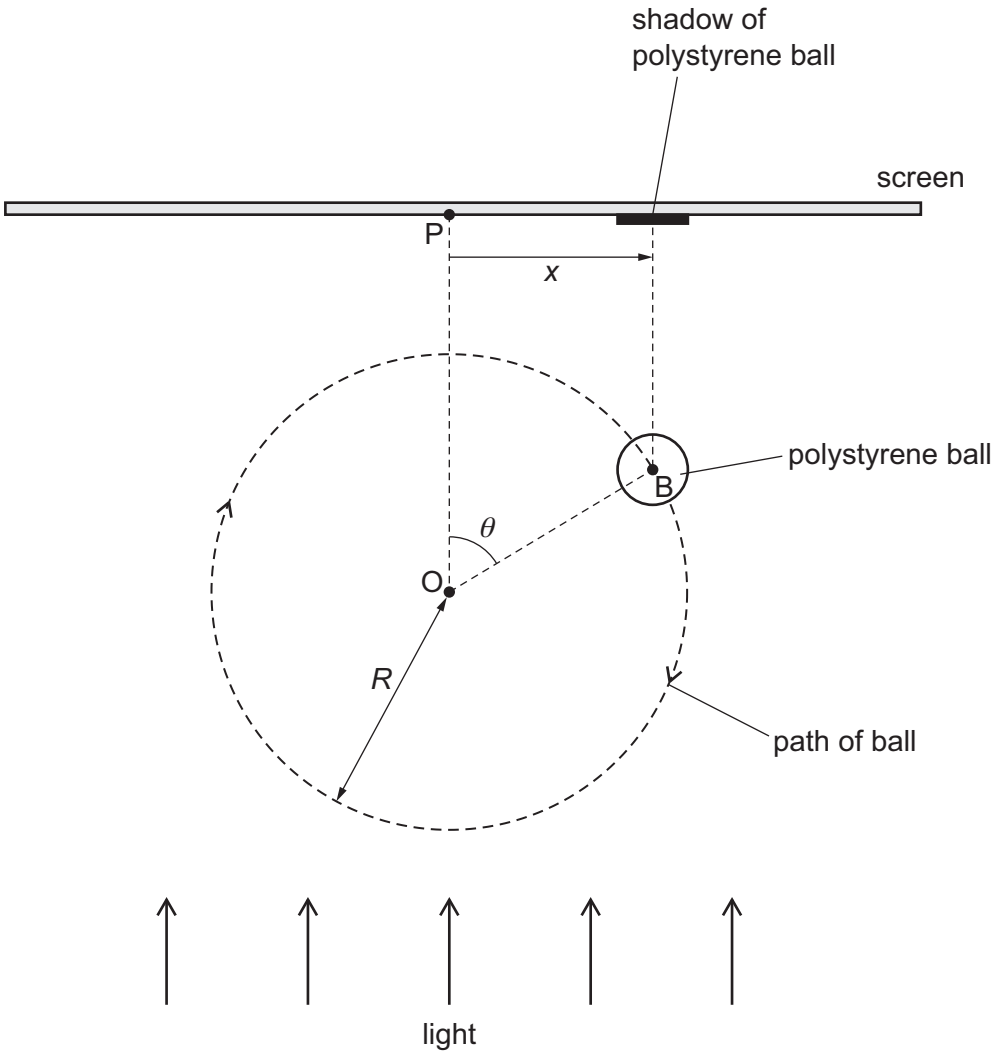


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω.

(i) State an expression, in terms of R and ω, for the speed v of the ball.

v = ... [1]

(ii) Determine an expression, in terms of v and ω, for the centripetal acceleration of the ball.

centripetal acceleration = ... [2]

(c) The ball in **(b)** is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

(i) Determine an expression, in terms of R and θ, for the displacement x of the shadow from P.

x = ... [1]

(ii) The value of θ is zero at time t = 0.

State an expression for θ in terms of ω and t.

θ = ... [1]

(iii) Use your answers in **(c)(i)** and **(c)(ii)** to show that x is given by

$$x = R\sin\omega t.$$

[1]

(iv) Explain, with reference to the equation in **(c)(iii)**, why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....

..... [1]

(d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1}.

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

(i) the amplitude

amplitude = ... m [1]

(ii) the period

period = ... s [2]

(iii) the maximum acceleration.

maximum acceleration = ... m s^{-2} [2]

(e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]

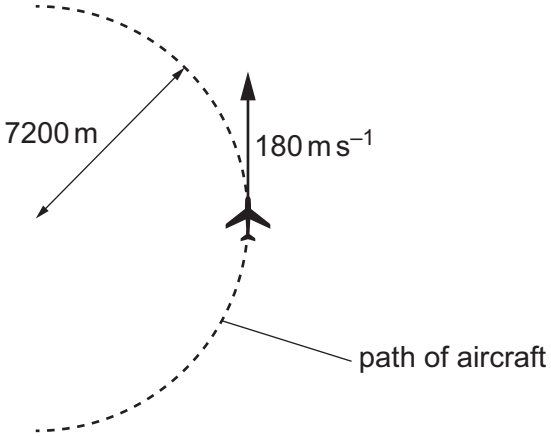
1 **(a)** With reference to velocity and acceleration, describe uniform circular motion.

.....

.....

..... [2]

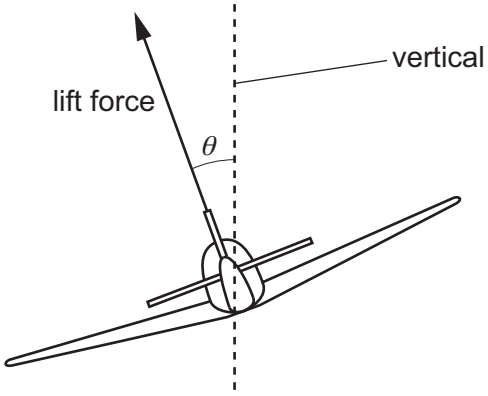
(b) An aircraft is travelling in a horizontal circle of radius 7200 m at a speed of 180 m s^{-1}, as shown in the view from above in Figure 1.1.



not to scale

Figure 1.1

In order to move in a circle, the aircraft has to tilt its wings so that the lift force from the wings is not vertical, as shown in the view from behind in Figure 1.2.



not to scale

Figure 1.2

(i) With reference to the forces acting on the aircraft, explain why the circular motion of the aircraft requires the lift force to be in the direction shown rather than vertically upwards.

.....

.....

..... [2]

(ii) On Figure 1.2, draw an arrow from the aircraft to represent the direction of the resultant force on the aircraft. [1]

(iii) Calculate the centripetal acceleration of the aircraft.

acceleration = ... $m\,s^{-2}$ [2]

(iv) Determine the angle θ at which the aircraft is tilted. Explain your reasoning.

θ = ... ° [3]

[Total: 10]

Question	Answer	Marks
1(a)	acceleration perpendicular to velocity	**B1**
1(b)(i)	decreases	**B1**
1(b)(ii)	(acceleration of) $9.8\,m\,s^{-2}$ is caused by weight of car **or** centripetal force must be greater than weight of car	**B1**
	(acceleration > $9.8\,m\,s^{-2}$) requires contact force from track **or** (centripetal force > weight) requires contact force from track	**B1**
1(c)	$\frac{1}{2}mv_Y^2 = \frac{1}{2}mv_X^2 - mgh$	**C1**
	$a = v^2 / r$	**C1**
	$v_Y^2 = 3.8^2 - 2 \times 9.81 \times 0.62$ so $v_Y = 1.5\,m\,s^{-1}$ $a = 1.5^2 / 0.31 = 7.3\,m\,s^{-2}$ (which is less than $9.8\,m\,s^{-2}$) so no	**A1**
	or	
	$v_Y = \sqrt{(9.81 \times 0.31)} = 1.74\,m\,s^{-1}$ so $v_X^2 = 1.74^2 + 2 \times 9.81 \times 0.62$ $v_X = 3.9\,m\,s^{-1}$ (which is greater than $3.8\,m\,s^{-1}$) so no	**(A1)**
1(d)	acceleration is independent of mass so makes no difference **or** mass cancels in the equation so makes no difference	**B1**

Question	Answer	Marks
2(a)	horizontal force on sphere causes centripetal acceleration	**B1**
	weight of sphere is (now) equal to vertical component of tension **or** horizontal and vertical components (of force) (now) combine to give greater tension (in spring)	**B1**
	greater tension in spring so greater extension of spring	**B1**
2(b)(i)	$r = 10.8 \times \sin 27° = 4.9\,\text{cm}$	**A1**
2(b)(ii)	$T \cos \theta = mg$ **or** $T \cos \theta = W$ **and** $W = mg$	**C1**
	$T \cos 27° = 0.29 \times 9.81$ leading to $T = 3.2\,\text{N}$	**A1**
2(b)(iii)	$\Delta T = 3.2 - (0.29 \times 9.81)$	**C1**
	$k = \Delta T / \Delta x$ $= [3.2 - (0.29 \times 9.81)] / [10.8 - 8.5]$ $= 0.15\,\text{N cm}^{-1}$	**A1**
2(c)(i)	centripetal acceleration $= (T \sin \theta) / m$ $= (3.2 \times \sin 27°) / 0.29$	**C1**
	$= 5.0\,\text{m s}^{-2}$	**A1**

Question	Answer	Marks
2(c)(ii)	$a = r\omega^2$ **and** $\omega = 2\pi / T$ **or** $a = v^2 / r$ **and** $v = 2\pi r / T$	**C1**
	$T = 2\pi \times \sqrt{(0.049 / 5.0)}$ $= 0.62$ s	**A1**

Question	Answer	Marks
1(a)(i)	radius = $6.37 \times 10^6 \times \cos 52.2° = 3.90 \times 10^6$ m	**A1**
1(a)(ii)	period = 24 hours	**C1**
	$v = 2\pi r / T$ **or** $v = r\omega$ **and** $\omega = 2\pi / T$	**C1**
	$v = (2\pi \times 3.90 \times 10^6) / (24 \times 60 \times 60)$ $= 280$ m s^{-1}	**A1**
1(b)(i)	$F = mv^2 / r$	**C1**
	$= (58.6 \times 280^2) / (3.90 \times 10^6)$ $= 1.2$ N	**A1**
1(b)(ii)	arrow pointing horizontally to the left	**B1**
1(b)(iii)	arrow from student pointing along the dotted line, labelled ‘weight’	**B1**
	upwards arrow from student pointing in a direction to the left of normal and above the tangent to the Earth, labelled ‘contact force’	**B1**

Question	Answer	Marks
1(a)	angle (subtended at centre of circle) when arc length = radius	**B1**
1(b)	$\omega = 2\pi / T$	**C1**
	$= 2\pi / (1.0 \times 60 \times 60)$ $= 1.7 \times 10^{-3}$ rad s^{-1}	**A1**
1(c)(i)	angle $= 1.7 \times 10^{-3} \times 1400$ $= 2.4$ rad	**A1**
1(c)(ii)	L = arc length / angle $= 0.44 / 2.4$ **or** $L = 0.44 \times (3600 / 1400) / 2\pi$	**C1**
	$L = 0.18$ m	**A1**
1(c)(iii)	$a = r\omega^2$	**C1**
	$= 0.18 \times (1.745 \times 10^{-3})^2$ $= 5.5 \times 10^{-7}$ m s^{-2}	**A1**
1(d)	centripetal acceleration is negligible compared with acceleration of free fall **or** numerical comparison establishing answer to **(c)(iii)** $\ll 9.81$	**B1**
	resultant force is negligible compared with weight (of modelling clay) (so variation is negligible) **or** force exerted by minute hand (approximately) equal (and opposite) to weight of modelling clay	**B1**

Question	Answer	Marks
1(a)	arrow vertically downwards labelled 'weight' and arrow perpendicular to cone, inwards and upwards, labelled 'normal contact force'	**B1**
1(b)	vertical component of contact force = weight (so no resultant force vertically)	**B1**
	horizontal component of contact force is resultant force towards centre (of circle)	**B1**
1(c)	$a = v^2 / r$	**C1**
	$a = g \tan 52°$	**C1**
	$9.81 \times \tan 52° = v^2 / 0.15$ leading to $v = 1.4\ \text{m s}^{-1}$	**A1**
1(d)	$v = r\omega$ or $a = r\omega^2$	**C1**
	$\omega = 1.4 / 0.15$ or $\sqrt{(9.81 \times \tan 52° / 0.15)}$ $= 9.3\ \text{rad s}^{-1}$	**A1**
1(e)	same resultant force / same acceleration so v^2 is proportional to r (so if speed increases radius must also increase)	**A1**

Question	Answer	Marks
1(a)	velocity and acceleration both have constant magnitude	**B1**
	velocity is (always) perpendicular to acceleration	**B1**
1(b)(i)	$v = R\omega$	**A1**
1(b)(ii)	$a = R\omega^2$ **or** $a = v^2 / R$	**C1**
	$a = v\omega$	**A1**
1(c)(i)	$x = R \sin \theta$	**A1**
1(c)(ii)	$\theta = \omega t$	**A1**
1(c)(iii)	clear substitution of $\theta = \omega t$ into $x = R \sin \theta$ leading to $x = R \sin \omega t$	**A1**
1(c)(iv)	equation is of the form $x = x_0 \sin \omega t$ (so simple harmonic motion)	**B1**
1(d)(i)	amplitude $= 0.46 / 2$ $= 0.23$ m	**A1**
1(d)(ii)	$\omega = 2\pi / T$	**C1**
	period $= 2\pi / 1.9$ $= 3.3$ s	**A1**

Question	Answer	Marks
1(d)(iii)	$a_0 = \omega^2 x_0$	**C1**
	$= 1.9^2 \times 0.23$ $= 0.83\ \text{m s}^{-2}$	**A1**
1(e)	shadow on screen, labelled A, above left-hand edge of the circular path	**B1**

Question	Answer	Marks	Guidance
1(a)	acceleration is (always) perpendicular to velocity	**B1**	
	velocity and acceleration have constant magnitude **or** velocity and acceleration change direction	**B1**	Accept in terms of either velocity or acceleration (both not needed). 'Constant speed', with no reference to magnitude of velocity, is insufficient. **Allow** 'acceleration is (always) directed towards the centre of the circle'. **Allow** 'velocity is (always) tangential to the circle'.
1(b)(i)	only other force is weight, which is vertical **or** weight has no horizontal component **or** vertical component of lift is equal (and opposite) to the weight	**B1**	**Allow** 'gravitational force' for 'weight'. **Allow** 'balances' or 'cancels' for 'equal to'.
	lift must have a horizontal component to cause centripetal acceleration	**B1**	**Allow** 'Horizontal component of lift provides / is the resultant / centripetal force'.
1(b)(ii)	arrow drawn from aircraft, horizontally to the left	**B1**	Arrow does not need to touch the aircraft but be pointing away from the aircraft and pass through some part of the aircraft when extended. **Allow** tolerance ± 5°. Treat any other arrows that (extended if necessary) pass through some part of the aircraft as contradictory unless labelling clearly conveys that one is the resultant force or the other arrows are not.
1(b)(iii)	$a = v^2 / r$	**C1**	**Allow** $a = \omega^2 r$ **and** $v = r\omega$.
	$a = 180^2 / 7200$ $= 4.5\ \text{m s}^{-2}$	**A1**	Correct to at least 2 significant figures. AFC.

Question	Answer	Marks	Guidance
1(b)(iv)	vertical component of lift balances the weight **or** horizontal component of lift causes the centripetal acceleration **or** $mg = L\cos\theta$ **and** $ma = L\sin\theta$	**B1**	**Allow** 'equals' for 'balances'. **Allow** 'provides … force' for 'causes … acceleration'. **Allow** *F* for *L*. **Allow** '*W*' for '*mg*' and 'F_C' or 'mv^2/r' for '*ma*'.
	$\tan\theta = a / g$	**C1**	
	$\tan\theta = 4.5 / 9.81$ $\theta = 25°$	**A1**	Possible ECF from **1(b)(iii)**. Correct to at least 2 significant figures. (24.6)